

FLOATING IN-OUT ROUTINES 2/24/67 (FINOUT,14)

/APR 8-64, MOD SRW 12-65

/FLIP 10T

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FLIP,      LAC I (FSA)
FLIP+1,    DAC A      /KEEP TEXT PTR IN A
           DCM G      /HIGH ORDER MANTISSA IN G
           DCM H      /LOW ORDER MANTISSA IN H
           DCM C      /CHARACTERISTIC IN C
           DCM I (ATEM) /CORRECTIVE SCALE FACTOR IN ATEM
           DCM B      /EXPONENT OF "E" IN B

CHLPLS,
FLIP1,     LCH I A      /MAIN LOOP, CLASSIFY CHARACTER
           SAD (CHARAC L,)
           JHP FLIP4
           SAS CHLPLS    /((CHARAC L+))
           SAD (CHARAC L-)
           JHP FLIP5
           SEA I
           JHP FLIP5
           SAD CHARLE    /((CHARAC LE))
           JHP FLIP6
           RAL 6S        /TEST FOR A DIGIT
           XQR B13       /((2))
           SUB CLP,
           SPA          /WAS CHARACTER A DIGIT?
           JHP FLIP7     /YES, GO PROCESS IT,
           SZF 4         /ANY DIGITS IN THIS FIELD?
           JHP FLIP13    /YES, GO EVALUATE FOREGOING STRING,
           JSP FLIPX     /EXIT SUBROUTINE, FAILURE
           JHP R1        /GIVE R1 TO INDICATE FAILURE

FLIP2,
FLIP3,     SZF I 1      /DECIMAL POINT, CHECK FOR REDUNDANCE
           SZF 2        /HAVE WE SEEN AN "E" ALREADY?
           JHP FLIP2    /YES, THIS CHARACTER FAILS,
           STF 1        /FLAG 1 INDICATES "SEEN A DECIMAL POINT"
           JHP FLIP1    /GO GET NEXT CHARACTER

FLIP4,
FLIP5,     SZF I 3      /SIGN, CHECK FOR REDUNDANCE
           SZF 1        /HAVE WE SEEN A DECIMAL POINT ALREADY?
           JHP FLIP2    /YES, THIS CHARACTER FAILS,
           SZF 4        /HAVE WE ALREADY SEEN A DIGIT?
           JHP FLIP13   /YES, THIS CHARACTER FAILS
           STF 3        /INDICATES "SEEN A SIGN" IN THIS FIELD
           SAS (CHARAC L-)
           JHP FLIP1    /NO, GET NEXT CHARACTER,
           SZF 2        /SET FLAG 6 IF WE HAVE SEEN AN E
           STF 6        /FLAG 6 INDICATES A NEGATIVE EXPONENT OF "E"
           SZF I 2      /IF FLAG 2 IS NOT SET, SET FLAG 5
           STF 5        /FLAG 5 INDICATES A NEGATIVE MANTISSA
           JHP FLIP1    /GET NEXT CHARACTER

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FLIP4,  SZF I 2      /FOUND AN E, CHECK FOR PEDUNDANCY
        SZF I 4      /HAS THE E PRECEDED ALL DIGITS?
        JMP FLIP2    /YES, THIS CHARACTER FAILS
        CLF 1        /INITIALIZE FOR A NEW FIELD
        STF 2        /FLAG 2 INDICATES "SEEN AN "E""
        CLF 3        /DONE SO A SIGN WON'T FAIL
        CLF 4
        JMP FLIP1    /GET NEXT CHARACTER

FLIP7,  STF 4        /FOUND A DIGIT
        ADD C10,     /GETS VALUE OF DIGIT IN AC
        DAC E        /SAVE DIGIT IN E TEMPORARILY
        SZF 2        /HAVE WE SEEN AN "E"
        JMP FLIP12   /YES, GO BUILD UP EXPONENT OF "E",
        JSP FPM10    /NO, BUILD UP MANTISSA
        JSP FPNORM    /PREPARE TO ADD IN THIS DIGIT
        DIO H
        LAW 4        /NEEDED TO FINISH MUL BY 10
        ADD C
        SUB F        /SUBTRACT DECREMENT LEFT BY NORMALIZE ROUTINE
        DAC C        /SAVE NEW CHARACTERISTIC
        LAC E        /MAKE THIS DIGIT INTO A MANTISSA
        RAP 55
        SCI,USZL I   /TEST FOR 0 & CLEAR IO
        JMP FLIP8    /PREVIOUS MANTISSA WAS NOT ZERO, GO AROUND,
        DAC G        /STORE CURRENT DIGIT AS MANTISSA
        LAW 4        /WITH 4 AS CHARACTERISTIC
        DAC C
        JMP FLIP11   /GO CHECK FOR DIGIT TO RIGHT OF DEC. PT.

FLIP8,  DAC E        /SAVE HI-ORD PART OF MANTISSA, LO-ORD=0
        LAW 4        /CHAR.=4, SET UP TO UN-NORMALIZE FOR ADDING
        DAC F

FLIP9,  SAD C        /UN-NORMALIZING LOOP, FINISHED WHEN C=F,
        JMP FLIP10   /SHIFT MANTISSA RIGHT 1 BIT
        LAC E
        SCP 15
        DAC E
        IDX F        /ADD 1 TO CHARACTERISTIC TO PRESERVE VALUE
        JMP FLIP9    /GO AROUND AGAIN

FLIP10, LAC E        /ADD THE TWO 36-BIT MANTISSAS TOGETHER
        SXP
        TAD H
        SWP
        TAD G
        SCP 15
        AND (377777) /NUMBERS ARE +, CORRECT POSSIBLE OVERFLOW
        DAC G        /SET UP NORMALIZE ROUTINE
        JSP FPNORM    /GO NORMALIZE
        DIO H        /SAVE LO-ORD MANTISSA, (MUST BE NON=0)
        IDX C        /OFFSET UN-NORMALIZING SHIFT
        SUB F        /SUBTRACT NORMALIZING DECREMENT
        DAC C        /STORE NEW CHARACTERISTIC

FLIP11, SZF 1        /ARE WE TO THE RIGHT OF THE DECIMAL POINT?
        IDX I (ATEM) /YES, IDX THE CORRECTIVE SCALE FACTOR
        JMP FLIP1    /GET NEXT CHARACTER

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FLIP12,  LAC B           /GET CURRENT EXPONENT OF E
          MUL C10,        /GET LOW-ORDER PART OF PRODUCT
          SWP             /OVERFLOWS IF LARGER THAN 17 BITS
          SNI+USMA I      /OVERFLOW
          JMP FLIP18
          SAR 1S          /MULTIPLY LEAVES THINGS SHIFTED LEFT A BIT
          ADD E           /ADD IN CURRENT DIGIT
          DAC B           /SAVE NEW EXPONENT OF "E"
          JMP FLIP1       /GET NEXT CHARACTER

FLIP13,  IDX I (USERPC)  /SUCCESS, SET UP P2
          LAC B           /GET EXPONENT OF "E"
          SZA+USZF 6      /RESTORE SIGN UNLESS 0
          CMA
          SUB I (ATEM)    /COMBINE IT WITH CORRECTIVE SCALE FACTOR
          DAC I (ATEM)    /POWER OF TEN TO TAKE NUMBER TO

FLIP14,  LAC I (ATEM)    /GET CORRECTIVE SCALING FACTOR
          SZA I           /END TEST, QUIT WHEN CORRECTION = 0
          JMP FLIP17      /EXIT LOOP, NUMBER IS NOW OK
          SPA            /MUST WE MUL OR DIV TO GET CORRECTION TO 0?
          JMP FLIP16      /GO ADD 1 TO SCALE FACTOR AND DIVIDE BY TEN
          SUB B17         /SUB (1) AND DIV BY 10
          DAC I (ATEM)
          JSP FPM10       /FPM10 MULTIPLIES MANTISSA IN G AND H BY TEN
          JSP FPNORM      /GO NORMALIZE
          DIO H           /SAVE LOW-ORDER MANTISSA IN H
          LAR 4           /ADD (4) TO CHARACTERISTIC IS PART OF MULTIPLY

FLIP15,  ADD C           /ADD IN PREVIOUS CHARACTERISTIC
          SUB F           /SUBTRACT DECREMENT LEFT BY NORMALIZE ROUTINE
          DAC C           /SAVE NEW CHARACTERISTIC
          S&L I           /TEST FOR A ZERO MANTISSA
          JMP FLIP14      /MANTISSA WAS NON-ZERO, GO AROUND AGAIN,
          JSP FLIPX       /MANTISSA WAS ZERO, PREPARE TO EXIT
          JMP FADD6       /GO PUT ZERO IN FLOATING AC AND RETURN TO USER

FLIP16,  IDX I (ATEM)    /ADD 1 TO SCALE FACTOR & DIV BY 10
          JSP FPD10       /DIVIDE MANTISSA IN G AND H BY TEN
          JSP FPNORM      /GO NORMALIZE
          DIO H           /SAVE LOW-ORDER MANTISSA
          LAR I 3         /DECREMENT CHARACTERISTIC FINISHES DIV BY 10
          JMP FLIP15      /COMPUTE NEW CHARACTERISTIC & LOOP AROUND

FLIP17,  LAC G           /NUMBER IS NOW OF PROPER MAGNITUDE
          LIO H           /GET MANTISSA
          S&F 5           /IF FLAG 5 IS SET, MANTISSA IS NEGATIVE
          CMA+UCM1
          DAC G           /SAVE MANTISSA WITH CORRECT SIGN
          DIO H
          JSP FLIPX       /PREPARE TO EXIT
          LIO H           /GET LOW-ORDER MANTISSA IN I.O.
          JMP FADD4       /GO PACK # INTO FAC & RETURN

FLIP18,  IDX I (USERPC)  /OVERFLOW
          JSP FLIPX       /PREPARE TO EXIT
          JMP FPOVFL      /GO TO THE FLOATING OVERFLOW ROUTINE

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FLIPX,   DAP D           /FLIP EXIT SUBROUTINE
         LAC CJ
         DIP D           /SAVE RETURN IN D
         LAW I 1         /GET PTR TO CHAR AND UNSTEP IT
         ADD A
         IQC
         IQC
         DAC I (FSA)     /STORE PTR IN FSA
         JMP D           /RETURN

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/FLOPE & FLOPF IOTS

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FLOP1,   LIO I (JMODE)  /CHECK FOR JBH
         SPI+USZF 6
         JMP R2
         LAC I (FAC+1)  /GET FL PNT #
         LIO I (FAC)
         SPI           /MAKE SURE MANTISSA IS POSITIVE
         CHA+UCM1+USTF 1 /NEGATIVE FLAG
         AND (777420)   /GET RID OF CHARACTERISTIC
         DIO G         /SAVE 36-BIT POSITIVE MANTISSA IN G AND H
         DAC H
         LAW I 1       /C CONTAINS EXPONENT OF 10 LESS 1
         DAC C
         LAC I (FAC+1) /GET CHARACTERISTIC
         RAP BS
         SAR 90
         SAR 10
FLOP2,   DAC E         /SAVE IT IN E
         SAN          /IS CHARACTERISTIC >0?
         JMP FLOP6     /YES, DIVIDE FL, PNT, # BY 10 & IDX C
         ADD (3)
         SGA          /TEST WHETHER NUMBER IS <=3
         JMP FLOP7     /EXIT, CHARACTERISTIC IS =3, =2, =1, OR 0.
FLOP2,   JSP FPM10     /MUL FL, PNT, # BY 10 & DECREMENT C
         JSP FPNORM    /NORMALIZE
         DIO H         /LO-ORD MANTISSA
         SZL I         /ZERO?
         JMP FLOP4     /NO, NON-ZERO MANTISSA. GO AROUND.
FLOP3,   DZM C         /GO PRINT A 0
         JMP FLOP10    /EXIT WITH C, G, AND H ALL PLUS ZERO
FLOP4,   LAW I 1       /SUBTRACT 1 FROM EXPONENT OF TEN
         ADD C
         DAC C
         LAW 4
FLOP5,   ADD E         /PART OF MUL BY 10
         SUB F         /ADD IN CHARACTERISTIC
         JSP F         /SUBTRACT DECREMENT LEFT BY NORMALIZE ROUTINE
         JMP FLOP1     /GO AROUND AGAIN

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FLOP6,   JSP FPO10      /DIVIDE NUMBER BY TEN
         JSP FPNORM     /NORMALIZE
         DIO H         /SAVE LOW ORDER HALF OF MANTISSA
         SZL          /WERE WE NORMALIZING PLUS ZERO?
         JMP FLOP3     /YES, EXIT, PRINT OUT A ZERO,
         IDY C         /ADD 1 TO EXPONENT OF TEN
         LAM I 3       /PART OF DIV BY 10
         JMP FLOP5     /COMPUTE NEW CHARACTERISTIC & GO AROUND AGAIN

FLOP7,   SZF 3         /HAVE WE BEEN HERE BEFORE?
         JMP FLOP8     /YES, GO ON TO NEXT PART OF ROUTINE,
         STF 3         /REMEMBER THAT WE'VE BEEN HERE
         JMP FLOP2     /SOME -3 CHARACTERISTICS CAN GO TO 0, TRY

FLOP8,   LIO H         /FP # IN E,C,H; UN-NORMALIZE G,H TO GET E=0
         LAC E
FLOP9,   SZA I         /FINISHED?
         JMP FLOP10    /EXIT
         LAC G
         SCR 15
         DAC G
         IDY E
         JMP FLOP9

FLOP10,  DIO H         /PUT LOW ORDER MANTISSA BACK IN H
         LAM ATEM
         DAC A         /A IS TEN STORAGE PTR
FLOP11,  JSP FPM10     /LOOP TO GENERATE 8 DIGITS
         LAC G         /HI-ORD RESULT OF MUL
         DAC B
         AND (17777)   /REMOVE BITS TO LEFT OF BIN PT
         SCL 4S        /PUT BIN PT BETWEEN BITS 0&1
         DAC G         /STORE NEW MANTISSA
         DIO H
         LAC B         /GET NEW DIGIT ALONE IN LO END OF AC
         RAL 5S
         AND (17)
         DAC I A       /STORE IN TEN AREA
         IDY A         /MAKE A POINT TO NEXT REGISTER
         SAS (ATEM+10) /QUIT AFTER 8 DIGITS
         JMP FLOP11    /GO AROUND AGAIN
         LAC I (STS)
         DAC G         /SET UP BYTE PTR FOR TEXT
         IDY I (USERPC) /GET PTR TO FORMAT WORD
         DZM E
         DAP E         /SAVE PTR
         LCH I E       /1ST BYTE OF FORMAT WD IS # OF COLUMNS
         RAL 6S
         IDA
         CMA
         DAC I (ATEM+10) /COLUMN COUNT WD
         LCH I E       /NEXT BYTE IS # OF SIGNIFICANT FIGURES
         RAL 6S
         DAC B
         LCH I E       /3RD BYTE IS # OF PLACES PAST DEC. PT.
         RAL 6S
         IDA
         DAC F         /1 * # OF SIG. FIGURES

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LAG C          /THIS SECTION CALCULATES WHICH DIGIT TO ROUND
SZF 2          /SKIP ON F FORMAT
CLA
ADD F
LIA
SUB B
SZM           /ROUND ON ACCURACY OR DIGITS?
LID B         /ACCURACY
LAT           /# OF CHARACTERS TO BE TYPED
SPA          /TEST FOR ROUND BEING TOO FAR LEFT
STF 4         /TOO FAR LEFT, PF4 PREVENTS ROUNDING
SUB B14       /(8,)
SMA          /TEST FOR ROUND BEING TOO FAR RIGHT
STF 4         /PREVENT ROUNDING IF SO
ADD (ATEM+10) /GETS PTR AT REG TO ROUND
DAC A         /SAVE PTR FOR LOOP AT FLOP12
DAC E         /SAVE PTR MORE PERMANENTLY FOR USE LATER
LAW I 5       /SUB (5) & TEST WHETHER +0=
ADD I 4
SZF I 4       /SKIP TO SUPPRESS ROUNDING
SPA          /SKIP TO ADD CARRY TO NEXT CHARACTER
JMP FLOP14    /QUIT ROUNDING

CHRLEQ,
FLOP12,      DZM I A      /CARRY LOOP ON ROUND=UP
LAW I 1      /UNSTEP TEM PTR
ADD A
CHARL4,      DAC A
SAS (ATEM-1) /TEST WHETHER ROUND CARRIED OFF THE LEFT END
JMP FLOP13   /OTHERWISE, GO AROUND
IDX I (ATEM) /ATEM WAS 0, MAKE IT 1
IDX C        /ADD 1 TO EXPONENT OF TEN
IDX E
JMP FLOP14   /EXIT FROM LOOP

CHARLE,
FLOP13,      IDX I A      /ADD CARRY FROM PREVIOUS DIGIT
SAD C10,     /DOES THIS CHARACTER CARRY LEFT NOW?
JMP FLOP12   /GO AROUND CARRY LOOP AGAIN
FLOP14,      CLI         /ROUNDOFF NOW COMPLETE, READY TO PRINT
SZF 1        /FLAG 1 SET FOR NEGATIVE NUMBER
LID (CHARAC L-)
IDX I (ATEM+10)
SZF I 5
JSP TYOIF
LAW I 2      /INITIALIZE D FOR STORING DIGITS
SZF I 2
SUB C
SMA
LAW I 1
DAC D        /D GETS IDX'ED FOR EACH DIGIT, STORE "," AT 0
LAC C        /INITIALIZE B
SZF I 2
SMA
LAW I 1
ADD (ATEM)
DAC B        /B ALWAYS POINTS TO NEXT DIGIT, AFTER IDX

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FLOP15,  IDX D          /LOOP FOR PRINTING
        SZA           /SKIP IF TIME TO PRINT DECIMAL POINT
        JMP FLOP16     /OTHERWISE, GO AROUND
        LID (CHARAC L,) /PRINT DEC PT
        JMP FLOP20
FLOP16,  SAA F          /ALL DONE PRINTING DIGITS?
        JMP FLOP21     /QUIT PRINTING DIGITS, EXIT FROM LOOP
        IDX B          /GET PTR TO NEXT DIGIT TO BE PRINTED
        SUB (ATEM)     /FIGURE OUT IF TOO FAR LEFT
        SNA           /TOO FAR LEFT?
        JMP FLOP18     /NO, GO AROUND,
FLOP17,  LAW 20         /YES, PRINT A 0
        JMP FLOP19
FLOP18,  LAC B          /FIGURE OUT IF TOO FAR RIGHT
        SUB E          /E IS PTR TO DIGIT BOUNDED AT
        SNA           /TOO FAR RIGHT?
        JMP FLOP17     /YES, PAST SIGNIFICANT FIGURES, PRINT A 0
        LAW 20         /NO, GET INTERNAL CODE FOR THIS DIGIT,
        ADD I B
FLOP19,  RGR 6S         /POSITION INTERNAL CODE FOR PRINTING
FLOP20,  IDX I (ATEM+10)
        SEF I 5
        JSP TYOIF
        JMP FLOP15     /GO AROUND LOOP AGAIN
FLOP21,  SEF I 2         /FINISHED WITH DIGITS, TEST FOR E FORMAT
        JMP FLOP22     /"E" FORMAT, GO AROUND,
        LAW 4
        ADD I (ATEM+10)
        DAC I (ATEM+10)
        SEF 5
        JMP FLOP22
        LID CHARLE     /((CHARAC LE) E FORMAT, PRINT THE E
        JSP TYOIF
        LAC C          /GET EXPONENT OF E, TEST SIGN, AND TYPE IT
        LID (CHARAC L=)
        SNA
        CLI+UCMA
        DAC A
        JSP TYOIF
        LAC A
        CLI+UCMA+USWP
        RCL 1S
        DIV C10,
05,      5             /CONSTANT STORED AFTER DIVIDE
        RJP 6S
        RCL 6S
        IOP C2020
        RUP 6S
        RGR 6S         /POSITION FIRST DIGIT
        JSP TYOIF
        JSP TYOIF

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FLOP22,  LAC G           /UPDATE PTR
        SZF I 6
        DAC I (STS)
        SZF 5
        JMP FLOP23      /RIGHT ADJUST
        LAC ML4
        SZF I 6
        DCH I 6
        JMP R1

FLOP24,  CLI
        JSP TYCIF
FLOP23,  ISP I (ATEM+10)
        JMP FLOP24
        CLF 5
        JMP FLOP14

/SUBR TO MUL A 36BIT +MANTISSA IN G,H BY 10 & SHIFT BIN PT 4 TO RIGHT
FPM10,  DAP D
        LAC CJ           /RETURN
        DIP D
        LAC H           /GET MANTISSA, WITH HALVES REVERSED
        LIO G
        AND (777760)     /REMOVE LAST 4 BITS TO KEEP SIGN +
        RCR 33           /UN=NORM,ING SHIFT TO PROTECT SIGN
        DIO G
        RAP 15
        MUL C5           /(5), MUL (5) & SHIFT 1
        DAC F
        DIO H
        LAR 5
        MUL G
        RCR 15
        LAT
        ADD F           /COMBINE THE TWO HALVES OF THE NEW MANTISSA
        DAC G
        LIO H           /LEAVES LO-ORD IN H & IO, HI-ORD IN G
        JMP D           /RETURN

/SUBR TO DIV A 36BIT +MANTISSA IN G,H BY 10 & SHIFT BIN PT 3 TO LEFT
FPD10,  DAP D
        LAC CJ
        DIP D
        LAC G           /SET UP FOR DIVIDE
        LIO H
        SCR 15           /SIC,
        DIV CHARL4       /(2400000)
C1849,, 1849,
        DAC G
        CLA+USWP
        DIV CHARL4       /(2400000)
C720,, 720,
        PAL 15
        DAC H
        LIA
        JMP D           /SAME EXIT FORMAT AS FPM10
                        /RETURN

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START

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